

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

1. Student information

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- **Assignment ID:** HW02
- **Assignment date:** 21/06/2026
- **AI tool(s) used:** Gemini
- **AI tool(s) used:**
 - ☒ Yes
 - ☐ No

2. Audit Table

Prompt + Tool	AI Output	Verdict	Reasoning (ISTQB)	Student Fix
AI tool: Gemini 3.1 Pro Time: 7:18am 17/6/2026 Prompt: See Appendix A	See Appendix A	INCOMPLETE	Domain testing and BVA step in almost testcase missing step enter confirm password even input have that field.	Adding confirm password field and step enter confirm password in testcase 2,3,4,5,6,7,8,9,10
AI tool: Gemini 3.1 Pro Time: 7:46am 17/6/2026 Prompt: See Appendix B	See Appendix B	INCOMPLETE	Missing very important step in the main flow í confirm the password	Adding step confirm password for testcase 11,12,13,14,15,16,17,18,19
AI tool: Gemini 3.1 Pro Time: 6:32pm 19/6/2026 Prompt: Continue with FR-10 -	See Appendix C	COMPLETE	Have analsis deeply and correctly	None

Prompt + Tool	AI Output	Verdict	Reasoning (ISTQB)	Student Fix
Trạng thái Đơn hàng (Order State Machine) step-by-step with the first prompt (Domain testing) and the second prompt (Boundary Value Analysis)				
AI tool: Gemini 3.1 Pro Time: 7:10pm 19/6/2026				
Prompt: Continue with FR-15 - Quản lý Sản phẩm (Product CRUD) step-by-step with the first prompt (Domain testing) and the second prompt (Boundary Value Analysis)				
	See Appendix D	COMPLETE	Have analsis deeply and correctly	None
AI tool: Gemini 3.1 Pro Time: 9:32pm 19/6/2026				
Prompt: Continue with FR-10 trên Mobile (Hủy đơn hàng trên App) step by step prompt1 (domain testing) and then wirh prompt2(boundary value analysis):				
	See Appendix E	INCOMPLETE	It hallucinate that this case have BVA unlike FR-10	Fixed: Boundary Value Analysis for this feature is none because FR-10 contains no range of values since we can not force BVA onto non-ordered discrete states

Appendix A

- Prompt:

Act as a Software Testing Expert (QA/QC Expert). I will provide you with 2 attached documents:

- System specification file (README.md - System Requirements Specification).
- Testing technique lecture file (S04_Domain Testing.pdf).

Your task is to apply the Domain Testing technique (focusing on Equivalence Partitioning) to the feature: FR-03 - Quên mật khẩu & Đặt lại mật khẩu (2 bước).

Please execute STEP-BY-STEP according to the following instructions:

****Step 1: Read and analyze the documents****

- Carefully read the specification of the feature FR-03 - Quên mật khẩu & Đặt lại mật khẩu (2 bước) in the README.md file.
- Carefully read the slide S04_Domain Testing.pdf, paying special attention to the "General Approach", "Guidelines" (slides 8-11), and the table presentation format.
- List the Input and Output variables of this feature.

****Step 2: Identify Equivalence Classes (EC)****

STRICTLY apply the Guidelines in the slide to divide the ECs, specifically:

- If it is a Range: Identify 1 Valid EC and 2 Invalid ECs.
- If it is a Set: Identify a Valid EC for each element and 1 Invalid EC.
- If it is a "Must be" condition: Identify 1 Valid EC and 1 Invalid EC.
- Present the results in a standard Markdown Table as shown in the slide:
| Condition | Valid Equivalence Classes | Invalid Equivalence Classes |

****Step 3: Assign IDs to the partitions****

- Assign an EC code to each partition (e.g., EC1, EC2, EC3...) for easy mapping.

****Step 4: Design Best Representative Test Cases****

Choose a representative value for each partition. Adhere to the test case selection principles in the slide:

- Valid test cases: Try to cover as many Valid ECs in a single test case as possible.
- Invalid test cases: Each test case MUST cover EXACTLY ONE Invalid EC (do not combine multiple errors into 1 test case).
- Present the Test Case suite in a table with the following MANDATORY FORMAT:
| TestID | Objective | Preconditions | Input | Test step | Expected Result | Actual Result | Verdict | Equivalence Classes |
(Note: Leave the Actual Result and Verdict columns temporarily blank).

Think step-by-step, present clearly and in English. Start with Step 1!

- Answer:

Here is the step-by-step application of the Domain Testing technique (Equivalence Partitioning) for the FR-03: Quên mật khẩu & Đặt lại mật khẩu feature, following the guidelines provided in the materials.

Step 1: Read and Analyze the Documents

Based on the system specification for FR-03 and the Domain Testing guidelines, we first need to identify the variables involved in this feature.

Input Variables:

- Email: User's registered email address (Step 1).
- OTP: The 6-digit one-time password sent to the email (Step 2).
- New Password (Mật khẩu mới): Must meet strong password criteria (Step 2).
- Confirm Password (Xác nhận mật khẩu mới): Must match the New Password (Step 2).

Output Variables:

- System Action (Step 1): Generate OTP, send email, and transition UI to Step 2 ("Bước 1 / 2" indicator).
- System Action (Step 2): Successfully update the password and navigate to the Login page, OR display an appropriate error message.

Step 2 & 3: Identify Equivalence Classes (EC) and Assign IDs

We will strictly apply the equivalence partitioning guidelines based on the input conditions (Range, Set, or "Must be").

Range: 1 Valid EC, 2 Invalid ECs. **Must be:** 1 Valid EC, 1 Invalid EC.

Condition	Valid Equivalence Classes	Invalid Equivalence Classes
Email: Registration Status (Must be)	EC1: Email is already registered in the system.	EC2: Email is not

registered in the system.OTP: Length & Format (Range/Must be)EC3: Exactly 6 numeric digits.EC4: Less than 6 digits.EC5: More than 6 digits or contains non-numeric characters.OTP: Validity (Must be)EC6: OTP matches the generated code for the requested email.EC7: OTP does not match or is used for a different email.New Pass: Length (Range)EC8: $8 \leq \text{Length} \leq \text{Max System Allowed}$.EC9: Length < 8 characters.EC10: Length > Max System Allowed (e.g., buffer overflow/DB limit).New Pass: Uppercase (Must be)EC11: Contains ≥ 1 uppercase letter.EC12: Contains 0 uppercase letters.New Pass: Lowercase (Must be)EC13: Contains ≥ 1 lowercase letter.EC14: Contains 0 lowercase letters.New Pass: Number (Must be)EC15: Contains ≥ 1 numeric digit.EC16: Contains 0 numeric digits.New Pass: Special Char (Must be)EC17: Contains ≥ 1 allowed special character (@, \$, !, %, *, ?, &).EC18: Contains 0 allowed special characters.Confirm Pass: Match (Must be)EC19: Exactly matches the "New Password" field.EC20: Does not match the "New Password" field.

Step 4: Design Best Representative Test CasesFollowing the testing strategy guidelines:We choose at least one test case from each equivalence class. For Valid classes, we combine them to cover as many valid equivalence classes as possible in a single test case. For Invalid classes, each test case must cover one and only one invalid class to isolate the cause of failure.

TestID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes	TC																																																										
FR03_01	Verify successful password reset with all valid inputs.	Account test@eshop.com exists.	Email: test@eshop.com OTP: 123456 (Correct) New Pass: Test1234! Confirm: Test1234!	1. Enter Email and submit. 2. Note the generated OTP. 3. Enter valid OTP, New Password, and Confirm Password. 4. Submit.	Password resets successfully. System redirects to Login page.	EC1, EC3, EC6, EC8, EC11, EC13, EC15, EC17, EC19	TC_FR03_02	Verify behavior with an unregistered email.	No account exists for the email.	Email: none@eshop.com	1. Enter unregistered Email. 2. Submit.	System displays an error message. Does not proceed to Step 2.	EC2	TC_FR03_03	Verify behavior when OTP is less than 6 digits.	Account exists. OTP requested.	OTP: 12345	New Pass: Test1234! Confirm: Test1234!	1. Enter 5-digit OTP. 2. Enter valid passwords. 3. Submit.	System rejects input. Error message regarding OTP length is shown.	EC1, EC4, EC8, EC11, EC13, EC15, EC17, EC19	TC_FR03_04	Verify behavior with incorrect/mismatched OTP.	Account exists. OTP requested.	OTP: 999999 (Incorrect) New Pass: Test1234! Confirm: Test1234!	1. Enter incorrect 6-digit OTP. 2. Enter valid passwords. 3. Submit.	System displays "Invalid OTP" error. Password is not changed.	EC1, EC3, EC7, EC8, EC11, EC13, EC15, EC17, EC19	TC_FR03_05	Verify password rejection when length is under 8 characters.	Account exists. OTP requested.	OTP: Correct	New Pass: Te1!abc Confirm: Te1!abc	1. Enter correct OTP. 2. Enter 7-character password. 3. Submit.	System displays error: Password must be at least 8 characters.	EC1, EC3, EC6, EC9, EC11, EC13, EC15, EC17, EC19	TC_FR03_06	Verify password rejection when missing an uppercase letter.	Account exists. OTP requested.	OTP: Correct	New Pass: test1234! Confirm: test1234!	1. Enter correct OTP. 2. Enter password with no uppercase. 3. Submit.	System displays error: Password requires an uppercase letter.	EC1, EC3, EC6, EC8, EC12, EC13, EC15, EC17, EC19	TC_FR03_07	Verify password rejection when missing a lowercase letter.	Account exists. OTP requested.	OTP: Correct	New Pass: TEST1234! Confirm: TEST1234!	1. Enter correct OTP. 2. Enter password with no lowercase. 3. Submit.	System displays error: Password requires a lowercase letter.	EC1, EC3, EC6, EC8, EC11, EC14, EC15, EC17, EC19	TC_FR03_08	Verify password rejection when missing a number.	Account exists. OTP requested.	OTP: Correct	New Pass: TestPass! Confirm: TestPass!	1. Enter correct OTP. 2. Enter password with no numbers. 3. Submit.	System displays error: Password requires a number.	EC1, EC3, EC6, EC8, EC11, EC13, EC16, EC17, EC19	TC_FR03_09	Verify password rejection when missing a special character.	Account exists. OTP requested.	OTP: Correct	New Pass: Test12345 Confirm: Test12345!	1. Enter correct OTP. 2. Enter password with no special char. 3. Submit.	System displays error: Password requires a special

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character.EC1, EC3, EC6, EC8, EC11, EC13, EC15, EC18, EC19TC_FR03_10Verify
behavior when Confirm Password does not match.Account exists. OTP requested.OTP:
CorrectNew Pass: Test1234!Confirm: Test1234@1. Enter correct OTP.2. Enter
mismatched passwords.3. Submit.System displays error: Passwords do not match.EC1,
EC3, EC6, EC8, EC11, EC13, EC15, EC17, EC20
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Appendix B

- Prompt:

Continue to act as a Software Testing Expert (QA/QC Expert). Based on the Equivalence Classes you just analyzed above, we will proceed to the next step: Boundary Value Analysis (BVA).

Please review the S04_Domain Testing.pdf slide again, paying attention to the "Boundary Value Analysis" section (slides 24-27).

Your task is to execute the following STEP-BY-STEP:

****Step 1: Filter and analyze Ordered Fields****

- From the previous EC list, filter out the Input/Output variables that are ranges, countable, or have length limits (e.g., integers, string lengths, quantities).
- For each variable, indicate the Lower Boundary (LB) and the Upper Boundary (UB).

****Step 2: Identify the Boundary Points to test****

- Apply the BVA principles from the slide; for each boundary, explicitly list the exact values: LB-1, LB, LB+1 and UB-1, UB, UB+1.
- Present the boundary analysis results in a Markdown table:
| Input Variable | Lower Boundary (LB) | LB-1, LB, LB+1 | Upper Boundary (UB) | UB-1, UB, UB+1 |

****Step 3: Design a comprehensive Test Case suite for BVA****

- Convert the discovered boundary points into specific Test Cases.
- Ensure all boundary areas are fully covered to avoid missing logic errors like sign confusion (e.g., using < instead of <=).
- Present the Test Case suite in a table with the following MANDATORY FORMAT:
| TestID | Objective | Preconditions | Input | Test step | Expected Result | Actual Result | Verdict | Equivalence Classes |
(Note: TestID must continue from the TestID list in the Domain Testing section. Leave the Actual Result and Verdict columns temporarily blank).

Think step-by-step and briefly explain why you chose those boundary values. Start with Step 1!

- Answer:

Here is the step-by-step execution of the Boundary Value Analysis (BVA) for the FR-03 feature, applying the principles from the provided testing materials.Step 1:

Filter and Analyze Ordered Fields

From the previous Equivalence Partitioning (EP) analysis, we look for input variables that involve ranges, lengths, or countable elements. In FR-03, there are two distinct ordered fields based on string length:

- OTP Length (Số lượng chữ số của OTP):** The specification strictly requires exactly 6 digits. This means the Lower Boundary (LB) and Upper Boundary (UB) converge at the exact same point. LB: 6 UB: 6
- New Password Length (Độ dài mật khẩu mới):** The specification mandates a minimum of 8 characters. While no explicit maximum length is stated in the requirements, a standard security/database constraint (e.g., varchar(255)) is typically applied in practice. As a QA expert, I will establish an assumed Upper Boundary of 255 to ensure comprehensive testing. LB: 8 UB: 255 (Assumed system limit)

Step 2: Identify the Boundary Points to Test

According to the Boundary Value Analysis methodology outlined in the lecture slide, we must test the exact boundaries along with the values immediately adjacent to them: LB-1, LB, LB+1 and UB-1, UB, UB+1. By pushing the inputs slightly out of bounds (LB-1, UB+1) and precisely on the bounds (LB, UB), we can effectively catch off-by-one errors (such as a developer using < instead of <=).

Input Variable	Lower Boundary (LB)	LB-1	LB	LB+1	Upper Boundary (UB)	UB-1	UB	UB+1
OTP Length (number of characters)	6	5	6	7	6	5	6	7
New Password Length (number of characters)	8	7	8	9	255	254	255	256

(Note: Since the OTP length has the same LB and UB, the boundary points overlap, meaning we only need to test lengths of 5, 6, and 7).

Step 3: Design a Comprehensive Test Case Suite for BVA

Continuing from the previous EP test suite, here are the boundary-focused test cases.

Important Test Design Note: To isolate the exact boundary being tested, all other input variables in these test cases (such as the password complexity requirements: uppercase, lowercase, number, special character) are intentionally kept valid. This ensures that if a test fails, it fails specifically because of the length boundary.

Test ID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes	
TC_FR03_11	Verify system rejects OTP with length exactly below the boundary (LB-1).	Account test@eshop.com exists.	OTP requested: 12345 (Length: 5)	New Pass: Test1234!1. Submit email.	2. Enter 5-digit OTP. 3. Enter valid new passwords. 4. Submit.	System rejects input. Error message indicates OTP must be 6 digits.	EC	4	
TC_FR03_12	Verify system accepts OTP with length exactly on the boundary (LB/UB).	Account test@eshop.com exists.	OTP requested: 123456 (Length: 6, Valid)	New Pass: Test1234!1. Submit email.	2. Enter correct 6-digit OTP. 3. Enter valid new passwords. 4. Submit.	Password resets successfully. System redirects to Login.	EC	3	
TC_FR03_13	Verify system rejects OTP with length exactly above the boundary (UB+1).	Account test@eshop.com exists.	OTP requested: 1234567 (Length: 7)	New Pass: Test1234!1. Submit email.	2. Enter 7-digit OTP. 3. Enter valid new passwords. 4. Submit.	System rejects input. Error message indicates OTP must be 6 digits.	EC	5	
TC_FR03_14	Verify system rejects password with length exactly below the boundary (LB-1).	Account test@eshop.com exists.	OTP requested: Correct 6-digit	New Pass: Te1!abc (Length: 7)	Confirm: Te1!abc1. Enter correct OTP.	2. Enter 7-character password (meeting all complexity rules). 3. Submit.	System rejects input. Error message: Password must be at least 8 characters.	EC	9
TC_FR03_15	Verify system accepts password with length exactly on the boundary (LB).	Account test@eshop.com exists.	OTP requested: Correct 6-digit	New Pass: Te1!abcd (Length: 8)	Confirm: Te1!abcd1. Enter correct OTP.	2. Enter 8-character password. 3. Submit.	Password resets successfully. System redirects to Login.	EC	8
TC_FR03_16	Verify system accepts password with length exactly above the boundary (LB+1).	Account test@eshop.com exists.	OTP requested: Correct 6-digit	New Pass: Te1!abcde (Length: 9)	Confirm: Te1!abcde1. Enter correct OTP.	2. Enter 9-character password. 3. Submit.	Password resets successfully. System redirects to Login.	EC	8
TC_FR03_17	Verify system accepts password with length exactly below the upper boundary (UB-1).	Account test@eshop.com exists.	OTP requested: OTP:						

Correct 6-digitNew Pass: 254-character string (Valid)Confirm: 254-character string1. Enter correct OTP.2. Enter 254-character password.3. Submit.Password resets successfully. System redirects to Login.EC8TC_FR03_18Verify system accepts password with length exactly on the upper boundary (UB).Account test@eshop.com exists. OTP requested.OTP: Correct 6-digitNew Pass: 255-character string (Valid)Confirm: 255-character string1. Enter correct OTP.2. Enter 255-character password.3. Submit.Password resets successfully. System redirects to Login.EC8TC_FR03_19Verify system rejects password with length exactly above the upper boundary (UB+1).Account test@eshop.com exists. OTP requested.OTP: Correct 6-digitNew Pass: 256-character stringConfirm: 256-character string1. Enter correct OTP.2. Enter 256-character password.3. Submit.System rejects input. Error message prevents system crash or truncation.EC10

Appendix C

- Answer:

Here is the step-by-step application of Domain Testing (Equivalence Partitioning) and Boundary Value Analysis (BVA) for the FR-10: Trạng thái Đơn hàng (Order State Machine) feature.

PART 1: DOMAIN TESTING (Equivalence Partitioning)

Step 1: Read and analyze the documents

Based on the FR-10 specification and the provided Domain Testing slide, we must first identify the variables that control the state machine.

Input Variables:

- Current State: The current status of the order (pending, confirmed, shipping, delivered, canceled).
- Actor/Role: The user triggering the action (Admin, User, Unauthenticated).
- Target State / Action: The requested transition (e.g., confirm, ship, deliver, cancel).

Output Variables:

- System Action: State is successfully updated, OR an appropriate error message is returned preventing the transition.

Step 2 & 3: Identify Equivalence Classes (EC) and Assign IDs

State machines rely heavily on "Set" conditions (enumerated lists) and "Must be" logic (business rules).

Condition

Valid Equivalence Classes	Invalid Equivalence Classes
Actor Role (Set)	EC1: Role is Admin.
EC2: Role is User.	EC3: Role is Unauthenticated / Missing Token.
Current: pending (Set)	EC4: Target is confirmed.
EC5: Target is canceled.	EC6: Target is shipping or delivered (Skipping states).
Current: confirmed (Set)	EC7: Target is shipping.
EC8: Target is canceled.	EC9: Target is pending or delivered.
Current: shipping (Set)	EC10: Target is delivered.
EC11: Target is canceled (Admin only).	EC12: Target is pending or confirmed.

Final States (Must be)

- (None - No transitions allowed from final states)
- EC13: Any transition from delivered.
- EC14: Any transition from canceled.

Business Logic: Cancel Restrictions (Must be)

- EC15: User cancels from pending or confirmed.
- EC16: Admin cancels from shipping.
- EC17: User attempts to cancel from shipping.

Step 4: Design Best Representative Test Cases

Note: Valid test cases combine valid ECs. Invalid test cases isolate EXACTLY ONE invalid EC.

Test ID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict
Equivalence Classes	TC_FR10_01	Verify Admin can confirm a pending order.	Order is pending. Logged in as Admin.	Target: confirmed	1. Navigate to Order details.	2. Click Confirm.	Order state changes to confirmed.
EC1, EC4	TC_FR10_02	Verify User can cancel a pending order.	Order is pending. Logged in as User (owner).	Target: canceled	1. Navigate to Order History.	2. Click Cancel.	Order state changes to canceled.
EC2, EC5, EC15	TC_FR10_03	Verify Admin can ship a confirmed order.	Order is confirmed. Logged in as Admin.	Target: shipping	1. Navigate to Order details.	2. Click Ship.	Order state changes to shipping.
EC1, EC7	TC_FR10_04	Verify					

User can cancel a confirmed order. Order is confirmed. Logged in as User. Target: canceled1. Navigate to Order History. 2. Click Cancel. Order state changes to canceled. EC2, EC8, EC15TC_FR10_05 Verify Admin can complete a shipping order. Order is shipping. Logged in as Admin. Target: delivered1. Navigate to Order details. 2. Click Complete/Deliver. Order state changes to delivered. EC1, EC10TC_FR10_06 Verify Admin can cancel a shipping order. Order is shipping. Logged in as Admin. Target: canceled1. Navigate to Order details. 2. Click Cancel. Order state changes to canceled. EC1, EC11, EC16TC_FR10_07 Verify User CANNOT cancel a shipping order. Order is shipping. Logged in as User. Target: canceled1. Attempt to send cancel request via API or UI bypass. System rejects transition with error message. EC2, EC11, EC17TC_FR10_08 Verify skipping states is prevented. Order is pending. Logged in as Admin. Target: shipping1. Attempt to force state to shipping via API. System rejects transition. EC1, EC6TC_FR10_09 Verify going backwards in state is prevented. Order is confirmed. Logged in as Admin. Target: pending1. Attempt to force state to pending via API. System rejects transition. EC1, EC9TC_FR10_10 Verify state is immutable once delivered. Order is delivered. Logged in as Admin. Target: canceled1. Attempt to change state via API/UI. System rejects transition (Final State rule). EC1, EC13TC_FR10_11 Verify state is immutable once canceled. Order is canceled. Logged in as Admin. Target: pending1. Attempt to change state via API/UI. System rejects transition (Final State rule). EC1, EC14

PART 2: BOUNDARY VALUE ANALYSIS (BVA) Step 1: Filter and analyze Ordered Fields Reviewing the slide materials, Boundary Value Analysis is strictly designed for ordered fields where variables specify a range of values (e.g., numbers, counts, string lengths). It targets boundaries because logical inequalities (<, <=, >, >=) are frequently mis-specified in code. Looking at FR-10, the variables involved are: Current_State: Enumeration/Set (pending, confirmed, etc.) Actor_Role: Enumeration/Set (Admin, User) Target_State: Enumeration/Set QA Expert Conclusion for Step 1: There are no ordered fields or ranges present in the State Machine logic. The inputs are discrete, categorical sets. Therefore, Boundary Value Analysis (BVA) is Not Applicable to FR-10. A core competency in software testing is knowing when to apply a specific technique. Attempting to force BVA onto non-ordered discrete states (like trying to find the "boundary" between pending and confirmed) would violate the mathematical premise of the technique outlined in the lecture slide. All logic control for FR-10 is fully and properly covered by the Equivalence Partitioning test cases generated in Part 1

Appendix D

- Answer:

Here is the step-by-step application of the Domain Testing technique (Equivalence Partitioning) for the FR-15 - Quản lý Sản phẩm (Product CRUD) feature, strictly following the provided instructions and documents. Step 1: Identify Input & Output Variables Based on the General Approach for Domain Testing and the system specification for FR-15, we first determine the variables. Inputs: Product Name (name): Must be provided (required) and have a maximum of 255 characters. Price (price): Must be provided, must be a number, and must be a positive value (> 0). Category (category_id): Must be provided and must be selected from an available list. Outputs: Success: Product is successfully added/updated in the system. Error: System returns an invalid input error message. Step 2 & Step 3: Identify

Equivalence Classes (EC) and Assign IDs We divide the domains into sub-domains (equivalence classes) based strictly on the provided Guidelines. Range Conditions: Identify 1 valid and 2 invalid classes. "Must be" Conditions: Identify 1 valid and 1 invalid class. Set Conditions: Identify valid classes for elements and 1 invalid class. ConditionValid Equivalence ClassesInvalid Equivalence Classes

Product Name: Length RangeEC1: $1 \leq \text{length} \leq 255$ EC2: $\text{length} < 1$ (Empty/Null)EC3: $\text{length} > 255$ Price: "Must be" a NumberEC4: Is a numberEC5: Not a number (e.g., characters, symbols)Price: Positive Value RangeEC6: $\text{Value} > 0$ EC7: $\text{Value} \leq 0$ Category: Set / "Must be"EC8: Selected ID exists in the available listEC9: Empty / Not selectedEC10: Selected ID does not exist in the available list

Step 4: Design Best Representative Test Cases Following the test case selection guidelines: For valid classes, we choose test cases to cover as many valid equivalence classes as possible. For invalid classes, we choose test cases so that each covers one and only one invalid class. This ensures that if the test fails, we know exactly which invalid input caused it.

Test ID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result
TC01	Verify adding product with all valid inputs	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 25000000 Category: "Smartphones" (Valid ID)	1. Navigate to Product management. 2. Enter Name, Price, and Category. 3. Click Submit.	Product is created successfully.	EC1, EC4, EC6, EC8
TC02	Verify adding product with empty name	Logged in as Admin with valid JWT	Name: "" (Empty) Price: 25000000 Category: "Smartphones"	1. Navigate to Product management. 2. Leave Name blank. 3. Fill valid Price and Category. 4. Click Submit.	Error message: Name is required.	EC2, EC4, EC6, EC8
TC03	Verify adding product with name exceeding max length	Logged in as Admin with valid JWT	Name: 256 characters string Price: 25000000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter Name with 256 characters. 3. Fill valid Price and Category. 4. Click Submit.	Error message: Name exceeds 255 characters.	EC3, EC4, EC6, EC8
TC04	Verify adding product with non-numeric price	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: "abc" Category: "Smartphones"	1. Navigate to Product management. 2. Fill valid Name and Category. 3. Enter "abc" in Price. 4. Click Submit.	Error message: Price must be a number.	EC1, EC5, EC8
TC05	Verify adding product with zero/negative price	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 0 Category: "Smartphones"	1. Navigate to Product management. 2. Fill valid Name and Category. 3. Enter 0 in Price. 4. Click Submit.	Error message: Price must be positive.	EC1, EC4, EC7, EC8
TC06	Verify adding product with no category selected	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 25000000 Category: None selected	1. Navigate to Product management. 2. Fill valid Name and Price. 3. Leave Category unselected. 4. Click Submit.	Error message: Category is required.	EC1, EC4, EC6, EC9
TC07	Verify adding product with non-existent category ID	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 25000000 Category: ID "99999"	1. Navigate to Product management. 2. Fill valid Name and Price. 3. Inject/Select a non-existent category ID. 4. Click Submit.	Error message: Invalid category.	EC1, EC4, EC6, EC10

Here is the step-by-step execution of Boundary Value Analysis (BVA) for the Product CRUD feature, continuing from the Equivalence Partitioning phase.

Step 1: Filter and analyze Ordered Fields According to BVA principles, we only apply this technique to ordered, countable fields or fields with specific length/value limits. Based on the previous Equivalence Classes, we filter out two variables that fit this criteria:

Product Name Length (Tên sản phẩm): The specification requires the name to be mandatory (minimum length > 0) and have a maximum of 255 characters.

Lower Boundary (LB): 1 character. Upper Boundary (UB): 255 characters.

Price (Giá): The price must be a positive number (> 0) and uses the "đ"

unit. Since the Vietnamese Đồng (VND) does not practically use decimals in this context, the smallest valid integer is 1. There is no defined upper limit in the specification. Lower Boundary (LB): 1 (₫). Upper Boundary (UB): Not explicitly specified (N/A). We will focus only on the lower boundary for this field. Step 2: Identify the Boundary Points to test We apply the Boundary Value Analysis approach by testing exactly at the boundaries and just outside them: LB-1, LB, LB+1 and UB-1, UB, UB+1. This catches common off-by-one errors (like using < instead of <=). Input Variable Lower Boundary (LB) LB-1, LB, LB+1 Upper Boundary (UB) UB-1, UB, UB+1 Product Name Length 10, 1, 225 254, 255, 256 Price Value 10, 1, 2N/AN/ANote on Price: LB-1 is 0, which directly tests the "số dương (> 0)" constraint. Step 3: Design a comprehensive Test Case suite for BVA Converting these boundary points into executable test cases guarantees that the edges of our equivalence partitions are thoroughly verified. (Note: Test IDs continue from the previous EC test suite. Some inputs like Name Length 0 and 256 overlap with our invalid EC tests, but they are explicitly listed here to demonstrate complete boundary coverage).

Test ID	Objective	Preconditions	Input	Test Step	Expected Result	Actual Result	Verdict	Equivalence Classes
TC08	Verify adding product with Name length = 0 (LB-1)	Logged in as Admin with valid JWT	Name: "" (0 chars) Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Leave Name blank. 3. Fill valid Price and Category. 4. Click Submit.	Product Name is required.		Fail	
TC09	Verify adding product with Name length = 1 (LB)	Logged in as Admin with valid JWT	Name: "A" (1 char) Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 1-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.		Pass	
TC10	Verify adding product with Name length = 2 (LB+1)	Logged in as Admin with valid JWT	Name: "A1" (2 chars) Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 2-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.		Pass	
TC11	Verify adding product with Name length = 254 (UB-1)	Logged in as Admin with valid JWT	Name: String of 254 valid characters Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 254-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.		Pass	
TC12	Verify adding product with Name length = 255 (UB)	Logged in as Admin with valid JWT	Name: String of 255 valid characters Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 255-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.		Pass	
TC13	Verify adding product with Name length = 256 (UB+1)	Logged in as Admin with valid JWT	Name: String of 256 valid characters Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 256-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Error message: Name exceeds 255 characters limit.		Fail	
TC14	Verify adding product with Price = 0 (LB-1)	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 0 Category: "Smartphones"	1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 0 in Price. 4. Click Submit.	Error message: Price must be a positive number (> 0).		Fail	
TC15	Verify adding product with Price = 1 (LB)	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 1 Category: "Smartphones"	1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 1 in Price. 4. Click Submit.	Product is created successfully.		Pass	
TC16	Verify adding product with Price = 2 (LB+1)	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 2 Category: "Smartphones"	1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 2 in Price. 4. Click Submit.	Product is created successfully.		Pass	

Appendix E

- Answer:

Here is the step-by-step application of Domain Testing and Boundary Value Analysis for the feature FR-10 & FR-20: Hủy đơn hàng trên App (Mobile Order Cancellation), following your instructions.

PHASE 1: Domain Testing (Equivalence Partitioning)

Step 1: Read and analyze the documents Based on the specification for FR-10 and FR-20, the Mobile App allows users to cancel orders. The success of this action depends entirely on the current state of the order in the State Machine.

Inputs: Current Order Status (status): The state of the order before cancellation. Actor (role): The user performing the action. In the context of the Mobile App (FR-20), this is the end-user ("User"), not the Admin.

Outputs: Success: The order status is updated to canceled. Error: The system rejects the action with an appropriate error message (e.g., state transition not allowed).

Step 2 & Step 3: Identify Equivalence Classes (EC) and Assign IDs According to the Guidelines for a "Set of input values" where elements are handled differently, we identify Valid classes for allowable states and Invalid classes for unallowable states. FR-20 strictly defines that a User can only cancel an order when it is pending or confirmed. FR-10 dictates that Users cannot cancel during shipping, and delivered / canceled are final states.

Condition	Valid Equivalence Classes	Invalid Equivalence Classes
Current Order Status (Set)	EC1: Status is pending EC2: Status is confirmed	EC3: Status is shipping (User restricted) EC4: Status is delivered (Final state) EC5: Status is canceled (Final state) EC6: Invalid/Unknown status (e.g., null, processing)

Step 4: Design Best Representative Test Cases Since we are dealing with a Set condition, each element represents a distinct partition. We will test each Valid EC and each Invalid EC individually to ensure the State Machine transitions are strictly enforced.

Test ID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict
EC1TC17	Verify User can cancel a "pending" order	Logged in as User on Mobile App. User has an order in pending state.	Order Status: pending	1. Navigate to Order History. 2. Select the pending order. 3. Tap "Cancel Order".	Order status changes to canceled.		Pass
EC1TC18	Verify User can cancel a "confirmed" order	Logged in as User on Mobile App. User has an order in confirmed state.	Order Status: confirmed	1. Navigate to Order History. 2. Select the confirmed order. 3. Tap "Cancel Order".	Order status changes to canceled.		Pass
EC2TC19	Verify User cannot cancel a "shipping" order	Logged in as User on Mobile App. User has an order in shipping state.	Order Status: shipping	1. Navigate to Order History. 2. Select the shipping order. 3. Attempt to cancel.	"Cancel" button is disabled/hidden, OR system returns error: "Cannot cancel shipping order".		Pass
EC3TC20	Verify User cannot cancel a "delivered" order	Logged in as User on Mobile App. User has an order in delivered state.	Order Status: delivered	1. Navigate to Order History. 2. Select the delivered order. 3. Attempt to cancel.	"Cancel" button is disabled/hidden, OR system returns error: "Order is already completed".		Pass
EC4TC21	Verify User cannot cancel an already "canceled" order	Logged in as User on Mobile App. User has an order in canceled state.	Order Status: canceled	1. Navigate to Order History. 2. Select the canceled order. 3. Attempt to cancel again.	"Cancel" button is disabled/hidden, OR system returns error: "Order is already canceled".		Pass
EC5TC22	Verify system handles unknown/invalid state via API	Logged in as User.	Intercept API request to cancel order.	Order Status: unknown_status	1. Send API request to cancel an order with an injected unknown_status.	System returns an invalid transition error message.	Pass

PHASE 2: Boundary Value Analysis (BVA)

Step 1: Filter and analyze Ordered Fields To apply BVA, we must look for input variables that are ordered, countable, or have physical/logical limits (like lengths, ranges, or numerical constraints). Looking at the inputs for the "Cancel Order" feature:

Current Order Status: This is a

Categorical/Nominal Set (pending, confirmed, shipping, etc.). There is no logical numerical order or continuous range here. You cannot calculate pending - 1 or confirmed + 1.Actor/Role: This is also a Categorical Set (User vs. Admin).QA Expert Conclusion for Steps 2 & 3:Based on the principles defined in the Domain Testing lecture, Boundary Value Analysis is NOT APPLICABLE to this specific feature. BVA requires a measurable boundary (e.g., \$1 <= x <= 999\$). Because the cancellation logic strictly relies on a discrete State Machine without any time limits, quantity limits, or length boundaries mentioned in FR-10 or FR-20, attempting to force BVA here would be theoretically incorrect. The Equivalence Partitioning test cases (TC17 - TC22) already provide 100% logic coverage for this specific requirement.

3. Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	5	100%
VALID (correct, accepted as-is)	2	40%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	3	60%

4. Conclusion — When should AI be used (or not)

Based on the audit of the five test design artifacts, AI demonstrated strong capabilities in structuring test case tables and deriving Equivalence Classes from clearly defined constraints, such as data ranges and logical state machines (e.g., FR-10 and FR-15). It shines as a high-speed drafting tool that effectively handles repetitive test design formatting.

However, AI failed when dealing with implicit user workflows. It completely missed the crucial "Confirm Password" step in the FR-03 main flow and struggled with contextual boundaries, initially causing confusion regarding the applicability of Boundary Value Analysis for non-ordered discrete states (FR-Mobile).

I recommendate that AI should be used as a "co-pilot" to outline initial partitions and generate boilerplate test suites. Human QA engineers must always meticulously review the outputs to inject missing business logic, correct workflow gaps, and ensure testing techniques are theoretically sound.

5. Mandatory Disclosure

The test cases for FR-03, FR-10, FR-15 and FR-Mobile were initially generated by Gemini 3.1 Pro; I reviewed and modified the test steps and expected results for FR-03, added the missing 'Confirm Password' edge cases; the AI Audit Table verdicts, reasoning, student fixes, and this conclusion were written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

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- **Date:** 21/06/2026
- **Signature:**



The image shows a handwritten signature in black ink on a light gray checkered background. The signature consists of the letters 'A' and 'N' in a stylized, cursive font, followed by a long horizontal line. Below the line, the name 'Nguyễn An Nghiệp' is written in a similar cursive script.